

KATYAYANI UPADHYAY

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Profile Summary

AI/ML and Data Science Engineer specializing in GenAI, Agentic AI, NLP, and Computer Vision with experience building autonomous multi-agent systems, RAG pipelines, and scalable ML workflows. Developed AI applications achieving **70% faster research synthesis** through LangGraph-based LLM orchestration and semantic retrieval. Built CNN-based speech and vision models delivering **80%+ classification accuracy** on datasets exceeding **50,000+ samples**.

Education

A.P.J. Abdul Kalam Technical University

2026

B.Tech in Computer Science (Artificial Intelligence & Machine Learning)

Lucknow, India

• CGPA: 8.9 / 10.0 | Relevant Coursework: Machine Learning, Deep Learning, NLP, DSA, Computer Vision

Work Experience

NIELIT (National Institute of Electronics & Information Technology)

August 2025 – February 2026

AI/ML Engineer Intern

Lucknow, India

- Accomplished **80%+ multi-class emotion classification accuracy** across 50,000+ audio samples, as measured by ROC-AUC and confusion matrix evaluation, by engineering a CNN pipeline with Python and Librosa-extracted MFCC features and optimising the signal preprocessing stage to cut misclassification rate by 15%.
- Drove **faster, data-driven decision-making for a 15-member ML team**, as measured by a 15% performance uplift on held-out test sets, by delivering cross-functional statistical model evaluation reports and collaborating with senior engineers to redesign the noise-reduction ETL pipeline.

Projects

Agentic AI Research Platform | *GenAI, FastAPI, Agentic AI, LangGraph, LLM Orchestration, RAG, Docker* [GitHub](#) | [Live](#)

- Engineered an **autonomous multi-agent AI research system** leveraging LangGraph orchestration, RAG, and semantic retrieval pipelines to generate context-aware research reports from **20+ real-time web sources**.
- Reduced research synthesis time by **70%** through parallelized AI-agent workflows integrating Tavily web search, Groq LLM inference, and HuggingFace embedding pipelines, while enabling scalable FastAPI-based deployment with Docker containerization.

Speech Emotion Recognition System | *Python, CNN, Librosa, Flask, MLflow, Docker*

[GitHub](#)

- Secured **sub-200ms real-time inference** for a 6-class emotion classifier by deploying a CNN-based model via a Flask API, with model versioning and experiment tracking using MLflow.
- Reduced feature dimensionality by **40%** without accuracy loss by engineering MFCC-based audio features through Librosa and applying statistical feature selection on a 10,000+ sample dataset.

Age Detection — Real-time Computer Vision System | *Python, CNN, OpenCV, Keras, Streamlit*

[GitHub](#)

- Enabled **78%+ age prediction accuracy** (MAE-based) on a 5,000-image test set by leveraging **transfer learning** with a pre-trained CNN and tuning model performance using Keras.
- Architected a **real-time webcam-based age detection system** with less than **150ms latency per frame** by integrating OpenCV video capture with the trained model and deploying via Streamlit.

Technical Skills

AI / ML: Deep Learning, ML, CNNs, Transfer Learning, NLP, Text Classification, TF-IDF, Hyperparameter Tuning

Data Science: Exploratory Data Analysis (EDA), Feature Engineering, MFCC

MLOps & Deployment: MLflow, Model Versioning, Batch Inference, Model Serving, ETL Pipelines, Flask, Streamlit

Libraries & Frameworks: TensorFlow, PyTorch, Scikit-learn, OpenCV, Librosa, NumPy, Pandas, NLTK, Seaborn

Programming: Python, SQL, Java

Tools & Cloud: Git, Jupyter, Google Colab, Power BI, Tableau, Docker, AWS EC2

Languages: English (Advanced), Hindi (Native), Spanish (Intermediate)

Certifications

Foundations of Deep Learning: Concepts and Applications (Elite) — NPTEL (IISc Bangalore)

2026

Python for Data Science (Elite + Silver) — NPTEL (IIT Madras)

2026

Introduction to Machine Learning (Elite) — NPTEL (IIT Madras)

2025

Programming in Java (Elite + Silver) — NPTEL (IIT Kharagpur)

2025

Leadership Experience

IEEE Student Branch, AIMT

January 2024 – April 2025

Chairperson

Lucknow, India

- Boosted **student engagement by 60%** by leading 50+ volunteers, organizing 20+ technical workshops and securing 30k in sponsorship through stakeholder collaboration.